



CIVIL MAINT. Department
SIMHADRI

Ref: SMTTP/Civil Maint/2022-23/O&MCivil/109306

Date:

17/05/2022(18:35:38)

Subject: Review of LMI Residual Life and Condition Assessment of RCC and Steel Structures w.r.t. LMI/OPS/CIVIL/013

The LMI of "Residual Life and Condition Assessment of RCC and Steel Structures" with Ref. LMI/OGN/OPS/CIVIL/013, Rev. No.02 was prepared and approved in the month of MAY'2020.

The above LMI has been reviewed as on date and revision 03 enclosed. No changes in LMI are incorporated.

Put up for approval please.

Submitted by:

पांडब दत्ता - वरिष्ठ प्रबंधक

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On: 17/05/2022(18:35:38) **Ref:** SMTTP/Civil Maint/2022-23/O&MCivil/109306

The LMI of "Residual Life and Condition Assessment of RCC and Steel Structures" with Ref. LMI/OGN/OPS/CIVIL/013, Rev. No.3 (attached) may please be approved for implementation.

Submitted by:

जयंती गोपी किरण - उप महाप्रबंधक

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On:18/05/2022(16:56:17) **Ref:**SMTTP/Civil Maint/2022-23/O&MCivil/109306

LMI may please be approved.

Submitted by:

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On:20/05/2022(19:21:59) **Ref:**SMTTP/Civil Maint/2022-23/O&MCivil/109306

Competent authority may approve the LMI reviewed

Submitted by:

झिमलि धर - अपर महाप्रबंधक

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On:21/05/2022(09:04:13) **Ref:**SMTTP/Civil Maint/2022-23/O&MCivil/109306

Please discuss this LMI, with undersigned.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:23/05/2022(19:48:26) **Ref:**SMTTP/Civil Maint/2022-23/O&MCivil/109306

Discussed. May please be forwarded for approval.

Submitted by:

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On:01/06/2022(10:57:47) **Ref:**SMTTP/Civil Maint/2022-23/O&MCivil/109306

The LMI revision may please be accorded approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:08/06/2022(18:45:37) **Ref:**SMTTP/Civil Maint/2022-23/O&MCivil/109306

कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

Submitted by:

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On:09/06/2022(20:10:34) **Ref:**SMTTP/Civil Maint/2022-23/O&MCivil/109306

Approved & Submitted by:

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On:10/06/2022(09:15:23) **Ref:**SMTTP/Civil Maint/2022-23/O&MCivil/109306

NTPC

SIMHADRI.

LOCATION MANAGEMENT INSTRUCTION

LMI /OGN/OPS/CIVIL/013/

Rev. No. : 03 May 2022

***LMI for Residual Life and Condition
Assessment of RCC and Steel
Structures***

APPROVED FOR

Implementation-----

GGM, SIMHADRI.

Date-----

LMI APPROVAL DOCUMENTS

1. TITLE OF LMI	<i>LMI for Residual Life and Condition Assessment of RCC and Steel Structures</i>
2.a) DOCUMENT CODE b) REVISION/ISSUE NO.&DATE	LMI/OPS/CIVIL/013/ ISSUE-1; REV.-3; May'2022
3. ORIGINAL/REF.DOCUMENT	COS-ISO-00-OGN/OPS/CIVIL/013 Rev.No.:0 December 2011
4. PRERPARED BY	NAME: Shri. Pandab Datta Sr. Mgr (O&M-Civil) SIGNATURE-----
5. CHECKED BY	NAME: Shri. Jayanthi Gopi Kiran DGM (O&M-Civil) SIGNATURE----- NAME: Shri. Asis Kumar Mukhopadhyay AGM (MM-Offsites & Civil) SIGNATURE-----
6. REVIEWED BY	NAME: Shri. Mathew Eipe Kovoov GM (Maint.) SIGNATURE----- NAME: Shri. Rajesh Kumar Jha GM (O&M) SIGNATURE-----
7. APPROVED BY	NAME: Shri. Girish Chandra Choukse GGM (Simhadri) SIGNATURE: -----
8. DATE OF ISSUE	
9. INDEX LIST	ENCLOSED.
10. DISTRIBUTION LIST	ENCLOSED.

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1.0 INTRODUCTION

NTPC-Simhadri is having four 500 MW units under operation. As the location of plant is proximity to sea and sea water is used for cooling and Ash water slurry purposes, reinforced concrete and steel structures of plant undergo degradation due to corrosion susceptibility, impact of environment such as weathering action, chemical attack, abrasion etc. To mitigate any risk of catastrophic failure, aged structures of coastal based plants require structural integrity assessment primarily on need basis.

2.0 SCOPE

The scope of this LMI is to detail the various tests to be carried out in concrete structures, steel structures and machine foundations thereby enhancing the functional life of Power plant by

- a) Identification of RCC and Steel Structures' health deficiencies covering all aging and degradation aspects viz: carbonation of concrete, corrosion of steel, continuous and cyclic loading of structure, environmental impacts, creep & shrinkage, wear & tear, concrete degradation, fire accidents etc.
- b) Based on the results of assessment studies an engineering solution to Repair/ Rebuild/Modify along with the basis thereof including in terms of cost economics with various options suggested for implementation.

3.0 REFERENCE DOCUMENT

Operation guidance note COS-ISO-00-OGN/OPS/CIVIL/013 circulated by operation services-NTPC Corporate center.

4.0 SUPERSEDED DOCUMENT

LMI/OPS/CIV/013 of Rev-01

5.0 PRACTICES TO BE FOLLOWED FOR CONDITION ASSESMENT

Sl.No	Location	Action	Frequency	Responsibility
	Concrete structures			
1	Ash handling buildings	VI,CST,RHT,CMT, UPVT,HCPT	Inspection report of LMI/OGN/OPS/CIV/007& LMI/OGN/OPS/CIV/005 to be reviewed every year	HOD (O&M-Civil)
2	WTP buildings			
3	NDCT & Chimney			
4	Track hopper & wagon tippler tunnels			

			All other tests are Need based depending on the above inspection report.		
5	Sea water Pump house & jetty				
	Machine Foundations				
1	TG deck foundation	VI,CST,RHT,CMT, UPVT, HCPT DYNAMIC ANALYSIS (Measurement of amplitude, frequency of vibrations, natural frequency of foundation system, signature analysis of block foundation /deck slab of TG,PA/FD/ID Fan etc), CUT AND PULL OUT CAPO TEST	Inspection report of LMI/OGN/OPS/ CIV/011 to be reviewed every year. All other tests are need based depending on the above inspection report.	HOD (O&M-Civil)	
2	BFP & TDBFP foundation				
3	Mill foundations				
4	PA fan foundations				
5	FD fan foundations				
6	ID fan foundations				
	Steel structures				
1	TG hall structures	Carry out various tests like VI, DPT/ MPI, TM, RT, UT	Inspection report of LMI/OGN/OPS/ CIV/007 & LMI/OGN/OPS/ CIV/005 to be reviewed every year All other tests are need based depending on the above inspection report	HOD (O&M-MM)	
2	EOT crane structures				
3	Boiler and Auxillaries				
4	Boiler supporting structure				
5	Boiler column foundation				
6	ESP and duct supporting structures				
7	Chimney shell liner				
8	CHP crusher house, conveyor galleries, Transfer points			HOD (O&M-CHP)	

Abbreviations:

CST Core sample test (including testing sample for crushing strength, pH, CI, SO₄, Carbonation etc.

CMT Cover Meter Test

DPT Dye Penetration Test

MPI Magnetic Particle Inspection

RHT Hammer Test

UPVT Ultrasonic pulse velocity test

TM Thickness Measurement

VI Visual Inspection

HCPT: Half Cell Potential Test

RT: Radiographic Test

UT: Ultrasonic Test

6.0 EXCEPTION REPORTING

The exceptions as per the inspection report of LMI/OGN/OPS/CIV/007, LMI/OGN/OPS/CIV/005, LMI/OGN/OPS/CIV/011 are to be reported as per the following format by O&M-Civil

Decription of exception	Area	Location of exception	Remarks

7.0 RESPONSIBILITIES

The responsibility of carrying out different tasks in relation to inspection and testing of the above systems as detailed in the LMI are with HOD's of Civil/MM/CHP.

8.0 RECORD KEEPING

The records to be maintained under this LMI include.

- Inspection reports of LMI/OGN/OPS/CIV/005, LMI/OGN/OPS/CIV/007 & LMI/OGN/OPS/CIV/011
- Exception report.
- Compliance report.

The above records are to be maintained by HODs of Civil/MM/CHP.

9.REVIEW

This document shall be reviewed by HOP, Simhadri on two yearly basis or as necessary.

[illegible]

DISTRIBUTION LIST

Sl.No	Date of issue	Issued to	Copy No
1	Head of O&M		
	Head of TS		
	AGM-CIVIL		
	AGM-MTP		
	STRUCTURAL COMMITTEE MEMBERS		

COMPLIANCE RECORD

STRUCTURE	ACTIVITY	COMPLIANCE